

NeMu: a comprehensive pipeline for accurate reconstruction of neutral mutation spectra from evolutionary data

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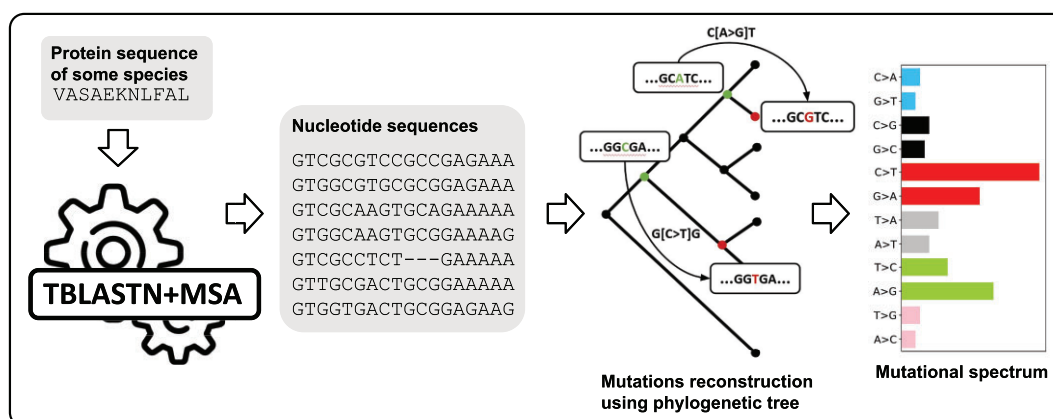
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Abstract

The recognized importance of mutational spectra in molecular evolution is yet to be fully exploited beyond human cancer studies and model organisms. The wealth of intraspecific polymorphism data in the GenBank repository, covering a broad spectrum of genes and species, presents an untapped opportunity for detailed mutational spectrum analysis. Existing methods fall short by ignoring intermediate substitutions on the inner branches of phylogenetic trees and lacking the capability for cross-species mutational comparisons. To address these challenges, we present the NeMu pipeline, available at <https://nemu-pipeline.com>, a tool grounded in phylogenetic principles designed to provide comprehensive and scalable analysis of mutational spectra. Utilizing extensive sequence data from numerous available genome projects, NeMu rapidly and accurately reconstructs the neutral mutational spectrum. This tool, facilitating the reconstruction of gene- and species-specific mutational spectra, contributes to a deeper understanding of evolutionary mechanisms across the broad spectrum of known species.

Graphical abstract



Introduction

Research across several well studied species, including humans, has consistently revealed that mutation rates significantly fluctuate across genomic regions, with pronounced differences depending on specific nucleotide contexts (1–5). Moreover, these context-specific mutational spectra show considerable variation among species, driven by the complex dynamics of DNA damage, repair mechanisms, and the fidelity of DNA polymerases (6,7). Recent findings have highlighted the significance of reconstructing both context- and species-specific mutational spectra, demonstrating that these spectra influence both neutral and adaptive nucleotide substitutions

(8–10). Understanding mutagenesis rules is crucial to study positive selection trends (11–13)—a key aspect in research on viruses and other pathogens. While the significance of mutational spectra is broadly recognized, the application of these insights to non-model species beyond human cancers is advancing at a slow pace, yet it remains highly anticipated.

Intraspecific polymorphism data, stored in the extensive GenBank repository and encompassing a vast array of genes and species (2,14,15), remains an underutilized resource for reconstructing mutational spectra. Existing approaches to reconstructing mutational spectra from intra-species polymorphisms predominantly rely on comparing derived

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variants with generalized ancestral sequences (2,16,17). However, these approaches are (i) not comprehensive, as they overlook all intermediate substitutions along the inner branches of the phylogenetic tree, resulting in the loss of crucial information and (ii) not scalable, lacking the applicability to inter-species comparisons. Here, we introduce the NeMu pipeline, which is based on a phylogenetic framework and designed to overcome these two primary limitations: comprehensiveness and scalability. NeMu offers a fast, user-friendly and flexible (adaptable to various research needs and user expertise levels) solution for reconstructing mutational spectra. It ranges from default settings for beginners to advanced options for experienced users, including tests for potential selection on synonymous sites. Our pipeline is dedicated to enhancing molecular evolution progress by providing a null model of mutagenesis for various genes and species, moving beyond simplistic assumptions in current models, and facilitating the identification of key gene- and species-specific mutagens.

Methods and results

The user-friendly interface of our NeMu pipeline, available at <https://nemu-pipeline.com>, allows for rapid derivation of the phylogenetically neutral mutational spectrum, usually in about 5–10 minutes for a 1000-nucleotide gene with roughly 100 sequences, presented in a nucleotide database. It features flexible input and output options, accepting a species name and protein sequence as primary inputs and delivering a 12-component mutational spectrum as the primary output. The pipeline's effectiveness has been confirmed by an extensive analysis of 10 000 mitochondrial genes using the MIDORI2 database (18).

The pipeline's results have already enabled the derivation of mtDNA mutational spectra for thousands of vertebrate species, uncovering correlations between species' life-history traits and their mutational spectra (19,20). Based on the pipeline's result we reconstructed a detailed 192-component mtDNA mutational spectrum for a typical vertebrate species by integrating mutations from a thousand vertebrate species. This analysis enabled us to uncover and elucidate fundamental aspects of mtDNA mutagenesis, including strong effects of ssDNA damage, deficiencies in base excision repair (BER), the absence of mismatch repair (MMR), and a potential mutational signature of gamma DNA polymerase (21). These proof-of-principle studies highlight the pipeline's utility, enabling and streamlining the process of extracting and examining species-specific mutational spectra, a rapidly expanding area of research.

Pipeline overview

Data sampling and DNA sequences preparation

There are two input modes available: (i) a single-protein, single-species sequence and (ii) a manual mode that handles multiple nucleotide sequence alignments.

When a single protein sequence is submitted, the pipeline adopts a two-step process: it first identifies orthologous sequences for a given species within the curated nucleotide databases via *tblastn* (22), and subsequently aligns the retrieved nucleotide sequences. We developed our *tblastn* results parsing method to increase the sample size of analyzed data from Genbank Nucleotide collection (*nt*) database and at the same time to prevent neutral mutation spectrum reconstruction

from being influenced by uncertainties in gene annotations and poorly functional characterization of rare and non-canonical exons in eukaryotic genes. To achieve this, we are permitted to analyze only DNA or mRNA sequences, which must have a minimum coverage of 50% for any target sequence by query protein and minimum identity of 80%, and that is not significantly contaminated by intron sequences. Additionally, we implemented hierarchical alignment pruning procedure: from the draft MAFFT v7.505 (23,24) nucleotide alignment, we cut regions enriched with indels (which frequently correspond to rare used exons) using *goalign* v0.3.6a software (25); we use a state-of-the-art MACSE v2.07 *alignSequences* tool (which is accounting for frameshifts and stop codons) (26) to align the remaining unique sequences fragments that correspond to canonical exons and to inform about stop-codons; the final stage involves pruning alignment regions that are enriched with indels and extracting the alignment that is free from duplicated sequences and stop-codons.

When handling multiple nucleotide sequences as the primary input, the pipeline proceeds to align them using the approach described in the previous paragraph. We strongly advise focusing solely on orthologous protein-coding sequences and carefully preparing codon alignments (refer to [Supplementary Note 1](#) for details). Manual mode can significantly broaden the primary objective of our pipeline enabling custom sequences analysis, such as results of mutational experiments, somatic mutation data, and cross-species or genus sequence comparisons. To prevent cross-species mutational spectra from being biased by intra-specific polymorphisms, users must create alignments with one sequence per species.

The Genbank *nt* database encompasses a lot of known species and genes, but many species-gene combinations have no polymorphic sequences, resulting in frequent unsuccessful queries in NeMu. To enhance success rates, it is crucial for users to judiciously select species-gene combinations. Ideally, choose genes that are frequently sequenced and serve as specific markers in ecological, evolutionary, or physiological studies. Furthermore, conducting a preliminary NCBI protein search using the gene name can reveal which species have multiple sequence hits, significantly boosting the chances of successful research outcomes.

Phylogeny reconstruction

The NeMu pipeline requires at least four unique sequences in the ingroup for analysis, but adding more sequences is advisable for better-quality results. To enhance the accuracy of neutral mutational spectra, we've incorporated several phylogenetic data processing steps and filters in our pipeline: (i) outgroup-based re-rooting of the phylogeny, (ii) identifying outgroup branches with near-zero lengths, (iii) pruning excessively long branches from a phylogenetic tree and (iv) masking conservative alignment sites to exclude gene sites influenced by potential selection and thus ensuring that the mutational spectra reconstruction is focused on the most neutral substitutions (for more details, see [Supplementary Note 1](#), [Supplementary Figure S1](#) and [Supplementary Table S1](#)). To reconstruct and analyze phylogeny, we used IQ-TREE 2.2.0 (27–30), to detect branching anomalies, we used TreeShrink (31), and to reroot trees, we used Newick tools 1.6.0 (32).

Reconstruction of the ancestral states probabilities

Following the inference of the phylogenetic tree, the pipeline progresses to the ancestral states reconstruction phase, also

performed by IQ-TREE2. To clarify the reconstructed states, we apply the ‘uncertainty principle’ (33), wherein we assign an uncertainty coefficient to the states of each node. This coefficient considers the depth of each node in the tree and linearly reduces the probability of reconstructed substitutions from the leaves towards the root. By incorporating this approach, we enhance the accuracy of our ancestral states probability estimates for nodes near the root (see [Supplementary Note 2](#), Appendix S1, [Supplementary Figures S2--S8](#)). As a result, we establish a probability distribution for the nucleotides at every site of each node within the tree.

Mutational spectrum inference

By carefully extracting ancestral states on a phylogenetic tree, we can reconstruct a mutation spectrum. Each extracted mutation has probability, calculated as probability of ancestral (parent node) context multiplied by probability of descendant (child node) context. These mutations are filtered according to the probability threshold (0.3 by default, see [Supplementary Note 2](#), [Supplementary Figure S6](#)), sampled according to tree location and mutation types, and aggregated to the mutational spectrum through normalization by the trinucleotides frequencies (see details in [Supplementary Note S3](#), [Supplementary Figure S9](#)). To select such mutations from trees, we developed the Pymutspec Python package available on PyPI. Utilizing this package (version 0.0.8) we collect all single nucleotide substitutions with trinucleotide context from the phylogenetic tree according to ancestral states probabilities.

The majority of evolutionary studies indicate that synonymous nucleotide substitutions may be used as a surrogate for neutral mutational spectra. However, there is considerable evidence that specific classes of synonymous nucleotide substitutions may be beneficial or harmful (34–36). So, we allow users to focus on synonymous nucleotide substitutions in highly variable codon alignment positions.

Web Server implementation

This website is free and open to all users and there is no login requirement. This website does not use cookies or collect personal information. The user interface of NeMu pipeline has been developed using Python library Streamlit. It establishes a connection with a remote Linux server to execute the pipeline, which is encapsulated in a Nextflow script (37). The pipeline relies on a diverse set of dependencies such as Python, Perl, R and various programs like IQ-TREE2, newick-utils, mvview, mafft and BLAST+. To simplify execution, ensure compatibility across different operating systems, and perform additional functionality with high computational demands (for instance, see [Supplementary Note 4](#), [Supplementary Figure S10](#)), all these dependencies have been packaged within a Singularity container (38). Consequently, users can enjoy the convenience of executing the pipeline independently on their preferred operating system.

Inputs

There are two ways to run the NeMu pipeline: using automatic and manual nucleotide sequences sampling. Automatic sampling requires pasting or loading a single protein sequence (for example, a GenBank reference sequence of any protein-coding gene), specifying the species name, selecting the nucleotide database, and choosing the genetic code as

the main inputs. The default database is Nucleotide collection (nt) (Genbank version 259, 2024-03-03). Considering the high abundance of mitochondrial genes in GenBank, we integrated databases of 13 protein-coding genes of mtDNA into the pipeline using MIDORI2, a reference database of *Eukaryota* mtDNA sequences (18). This sampling mode is available only for the automatic species-specific spectra reconstruction.

The manual sampling type offers users more freedom to analyze a broad range of genes and taxa, even enabling the derivation of a mutational spectrum for an entire genus or family. To engage this mode, users need to select ‘Nucleotide Seqs’ as the sequences type, upload a FASTA file (no more than 2MB) with multiple nucleotide sequences (alignments are preferred but not required), choose the level of analysis (species-specific or comparative), and provide the name of the outgroup sequence included in the FASTA file. It’s important to note that this method is suited for nucleotide sequences that are clearly orthologous, as significant evolutionary inhomogeneity in non-orthologous sequences can negatively affect the results.

The advanced parameters of the pipeline consist of three distinct sections of settings, each altering different processes within the pipeline. (i) In phylogenetic inference, users can select their preferred substitution models that IQ-TREE2 utilizes for phylogenetic tasks. Additionally, users can modify the parameters of outlier branch filtration performed by the TreeShrink package. (ii) During the mutation extraction phase, users can choose to either ignore substitution probabilities or adjust the probability consideration thresholds. Users also have the option to enable or disable the uncertainty coefficient. Moreover, users are able to adjust the parameters related to the utilization of site rates—whether to mask highly constrained positions in the alignment or not. (iii) In the spectra calculation step, users can opt for specific mutations: synonymous, fourfold synonymous, or all, including nonsynonymous (which can be biologically meaningful for example in cancer data or in mutation accumulation experiments). They can also choose to analyze mutations from only terminal or internal branches. Note that the default spectrum on synonymous substitutions extracted from the entire tree will be calculated in any successful run.

Outputs

As the main output user can see several pages with spectra barplots (12 and 192 components), tables (12 and 192 components), a list of all reconstructed mutations, the reconstructed tree and the list of executed tasks. The reconstructed spectra of the pipeline are derived from the coding strand of DNA, corresponding to mRNA. This approach doubles the number of components from 6 to 12 and from 96 to 192, significantly improving our ability to investigate mutational or reparational asymmetries observed in nuclear, mitochondrial, bacterial, and viral genes. If necessary, users can merge complementary substitutions, like C > T and G > A, by averaging their rates, thereby reducing the component count from 12 back to 6 and from 192 back to 96.

Upon completion of the pipeline execution, users can access and download their results from the output page for up to 7 days. Users who provide their email will also receive a link to the output page and output archive attached. The output page features interactive barplots and tables of mutational spectra, details of reconstructed mutations, a summary of completed

tasks, and a download option. The output archive includes figures of spectra and all tables with reconstructed mutations, expected mutations used in spectrum normalization, and spectra. If the user specified in advanced pipeline parameters different mutation samples, there may be several output spectra files, e.g. only for synonymous mutations on the entire tree, for all mutations on internal tree branches, etc. Additionally, intermediate outputs including log files, nucleotide alignment with encoded headers, headers decoding file, phylogenetic tree, and site rates table are also presented. This allows users to review, trace back and, if desired, manually rerun steps with custom settings. All this detailed information—including high-quality codon alignments, trees rooted with outgroup data, and reconstructed ancestral states at each node—can be used to derive additional metrics such as dN/dS ratios, population polymorphisms and more, using other software.

Mutational spectra quality control

In order to ensure accuracy of the mutational spectra reconstructed by the NeMu pipeline, we simulated birth-death phylogenetic trees and alignments that matched them. To do it we employed the AliSim module of IQ-TREE 2.2.2.7 (39), with evolution parameters finely tuned to include the substitution model and its parameters (i.e. the base of expected mutational spectrum), tree size, nucleotide frequencies, and the proportion of invariant sites. Upon generating 800 replicates (see details in [Supplementary Note 5](#)), we executed our pipeline on this dataset to calculate the synonymous mutational spectra. Comparative analysis, utilizing cosine similarity as a metric, revealed a striking congruence between the expected (simulated) spectra and observed, reconstructed by the NeMu pipeline, as illustrated in [Figure 1](#). This high degree of consistency was maintained even under scenarios of extreme nucleotide evolution (e.g. far from nucleotide compositional equilibrium) characterized by totally random parameters. Such compelling evidence underscores the NeMu pipeline's robustness and its capability to yield precise analyses for a diverse array of species-specific neutral DNA evolutionary scenarios.

To additionally test the robustness of our pipeline, for our simulations we utilized the tree topology of 4705 mammalian species (40), which features a non-uniform distribution of branch lengths, in contrast to the homogenous branches in randomly generated birth-death trees. We produced 200 replicates with simulation parameters similar to those previously mentioned (detailed in [Supplementary Note 5](#), [Supplementary Figure S11](#)). For the comparative analysis, we applied Euclidean distance to measure the precision of differences between the simulated and reconstructed mutational spectra for each branch. To establish benchmarks for comparison, we computed Euclidean distances between pairs of simulated mutational spectra across replicates sharing the same model and root sequence type but varying in model parameters. [Figure 2](#) (and [Supplementary Figure S12](#)) displays the Euclidean distance distributions for mutational spectra derived from synonymous mutations. We observed that, in contrast to the reference neutral distributions, the Euclidean distances for compared spectra are significantly smaller ([Supplementary Tables S2 and S3](#)), suggesting a high degree of accuracy in the reconstructed branch-specific spectra. Notably, the data revealed that the most variable site class (cat6) exhibits the broadest overlap between the neutrally expected and the reconstructed-

simulated Euclidean distance distributions. This observation confirms the findings in [Supplementary Note 1](#), indicating that under natural conditions, only a subset of the substitutions in the most variable site classes may be regarded as likely neutrally evolved.

Star-like or caterpillar tree topologies, which arise from frequent gene mixing during recombination and/or from intensive recurrent *de novo* mutations, make tracking changes over time challenging (polymorphic scenario). To address this, the 'majority rule consensus', which focuses on deviations from the most common genetic variants, can be used to reconstruct the mutational spectrum (hereafter referred to as the majority-rule-consensus-based mutational spectrum). Conversely, trees with long inner branches suggest that genes have been passed down with little mixing (high signal of divergent evolution) and/or rare recurrent *de novo* mutations (phylogenetic scenario). This situation requires probabilistic reconstruction of ancestral states using a phylogenetic approach (hereafter referred to as the standard mutational spectrum). To distinguish between phylogenetic and polymorphic scenarios, we can analyze the properties of phylogenetic trees, specifically the ratio of the total length of the upper branches, U , to that of the inner branches, I . A ratio where U significantly exceeds I ($U \gg I$) indicates a polymorphic scenario (41). For each species in our dataset (7017 intra-species mitochondrial subsets), we estimated both majority-rule-consensus-based and standard mutational spectra, along with a metric that assesses the similarity of the phylogeny to a star-like one. We correlate the cosine similarity of majority-rule-consensus-based and standard spectra to normalized differences of tree lengths of star-like and native trees (for details, see [Supplementary Note 6](#)). As a result ([Supplementary Note 6](#), [Supplementary Figures S3--S14](#)), we show: (1) the significant distance between the lengths of star-like trees and native trees due to $U \sim I$ (the vast majority of mtDNA have no possibility of recombination due to strict maternal inheritance), and (2) the fact that the cosine similarity of majority-rule-consensus-based and standard mutational spectra are negative correlate (with exceptionally high statistical significance, $P < 10^{-20}$) with the difference in lengths between star-like trees and native trees. In other words, our results indicate that the cosine similarity of majority-rule-consensus-based and standard mutational spectra increases as the difference in length between star-like trees and native trees decreases. Our tests confirm the worth and meaning of reconstructing standard (phylogeny-based) mutation spectra for at least 92% of intra-species mitochondrial datasets ([Supplementary Note 6](#)). Additionally, by this relation we demonstrate the interchangeability of majority-rule-consensus-based and standard mutational spectra when gene intermixing signals are present (and/or recurrent *de novo* mutations present) and when sufficient mutational data is available for reconstruction of mutation spectra. Of course, there were specific cases and genomic contexts when the mutation rate changed significantly (e.g. a genome-wide mutation rate model Roulette take this into account (42)) and, as a consequence, recurrent *de novo* mutations occur frequently (4,42), however these cases are uncommon, slow and average mutating sites are consistent with the infinite-sites assumption (43,44). Taken together, the strategy of phylogeny-based (we called, standard) mutational spectra reconstruction that is based on the infinite-sites assumption can be considered as a promising approximation, especially in slowly evolving genome regions.

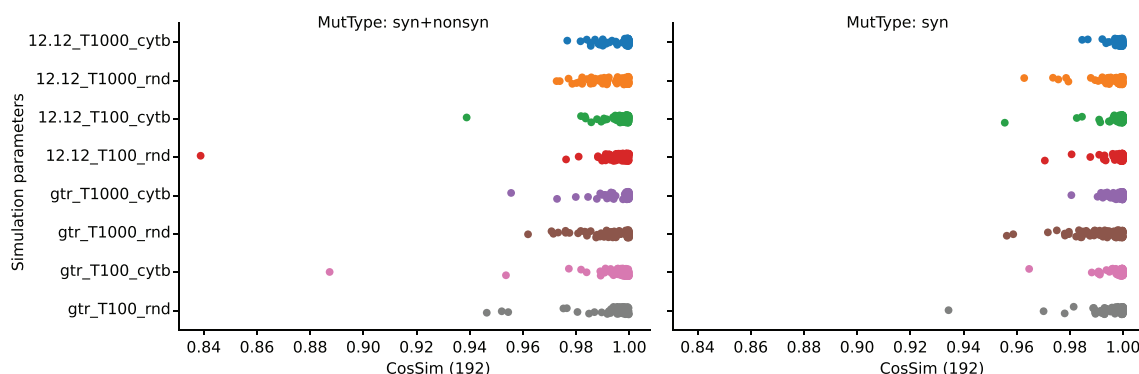


Figure 1. Testing for ancestral states and mutational spectra computational accuracy on random trees: the comparison between simulated and reconstructed 192-component mutational spectra on full trees. Y-axis indicates different simulation parameters: model (GTR or 12.12), tree size (100 or 1000 ingroup leaves) and root sequence (random or human CYTB gene). Each of eight parameter sets presented by 100 replicates.

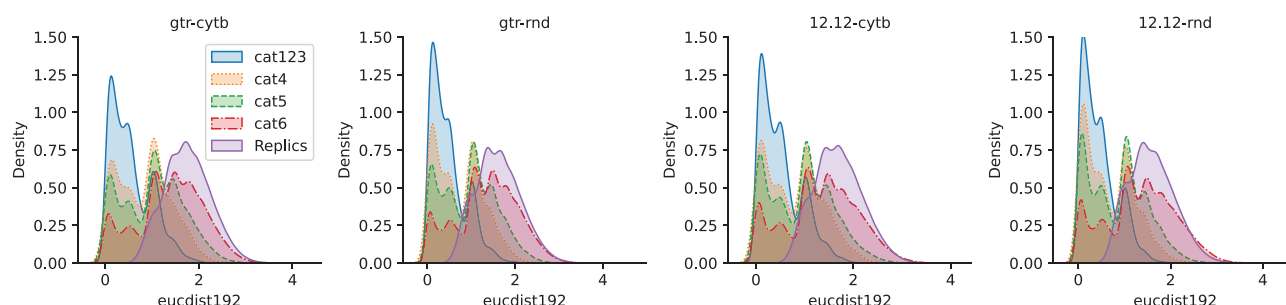


Figure 2. Testing for ancestral states and mutational spectra computational accuracy on a real tree of 4705 mammalian species: the comparison between simulated and reconstructed 192-component mutational branch-specific spectra. Different color densities indicate mutational spectra, calculated on sites with specified category of rate (cat123 for sites with 1, 2 and 3 rate category, cat4 for sites with 4 rate category, etc.). Violet color (Replicas) indicates reference distributions (pairwise Euclidean distances of simulated mutational spectra across replicates) to compare other distributions with. Four density plots show results for different simulation parameters: model (GTR or 12.12) and root sequence (random or human CYTB gene). Each of four parameter sets presented by 50 replicates.

Mutational spectra inference based on a large inter-species mtDNA dataset

To demonstrate the efficiency of our pipeline also on an inter-species scale with simultaneous analysis of hundreds of species, we used a challenging example - reconstructing mutational spectra of mitochondrial genes for more than 1300 mammalian species (Supplementary Files 1 and 2). It has been shown that the most neutral cancer-derived mtDNA mutational spectrum (45) is highly non-uniform with a strong excess of transitions, among which $G > A$ and $T > C$ are the most common ones. This mutational asymmetry is most likely driven by an excess of $C > T$ and $A > G$ on the heavy chain (correspondingly $G > A$ and $T > C$ on a light chain) due to its single-stranded mode during asynchronous replication (45–47). This mutational feature has been demonstrated to require non-reversible Markov models for analysis (46). The substantial divergence among the species analyzed, coupled with the anticipated non-uniform mutational spectrum, makes accurate ancestral state reconstruction not only particularly challenging but also critically important, a process we successfully demonstrated through this specific example. We show that both approaches to reconstruct ancestral sequences: PASTML (48) with MPPA time-reversible Markov models and the Brier score aware reconstruction based on HKY model, and our approach used in NeMu based on IQ-TREE2 and uncertainty principle (both GTR time reversible Markov model and RY10.12 (49), the best fitted

non-reversing Markov model) lead to very similar and anticipated results (Supplementary Figure S15 and S16): (i) uncover strong asymmetry of transitions ($G \rightarrow A \gg C \rightarrow T$, $T \rightarrow C \gg A \rightarrow G$), (ii) deficit of transversion over transitions, (iii) uncover asymmetry even between complementary transversions ($C \rightarrow A \gg G \rightarrow T$, $G \rightarrow C > C \rightarrow G$, $A \rightarrow C \gg T \rightarrow G$). We also demonstrated the robustness of interspecies mutational spectra to various mutation normalization schemas, as well as to time-reversible/time-non-reversible substitution model interchange. This data shows that our mutation reconstruction pipeline is both fast and accurate, even at the most challenging interspecies analysis level (see Supplementary Note 2, including Supplementary Figures S2–S3, and Supplementary Note 7).

Case studies

Mutational spectrum of genes exposed to prolonged time being single stranded

Another interesting feature of mtDNA mutagenesis is the $G > A$ substitution gradient, linked to the time being single-stranded (TBSS) due to asynchronous replication (50). This phenomenon suggests an increase in $G > A$ substitution asymmetry, measured as the $G > A/C > T$ ratio, especially in genes with prolonged TBSS. To test it with our pipeline, we quickly reconstructed mutational spectra of *Mus musculus* mtDNA genes: COX1 (with minimal TBSS) and CYTB (with maximal TBSS). Outputs for both

genes are permanently available for educational purposes (CYTB: https://nemu-pipeline.com/Output/?job_id=mouse_cytb; COX1: https://nemu-pipeline.com/Output/?job_id=mouse_cox1) and demonstrate expectedly higher asymmetry in CYTB compared to COX1 (CYTB: $G > A = 0.45$, $C > T = 0.11$, $G > A/C > T = 4.09$; COX1: $G > A = 0.32$, $C > T = 0.19$, $G > A/C > T = 1.68$). Further research with more genes and species could explore TBSS's impact on mutational spectra: whether the gradient differs between species; if there are other mutations affected by TBSS; what is the 192-component mutational signature of TBSS; etc.

Comparative analysis of mutational spectra in species with diverse life-history traits

One of the most straightforward applications of our pipeline is the reconstruction and comparison of species-specific mutational spectra. For instance, previously we have shown that $T > C$ substitutions in mtDNA ($A > G$ on a heavy chain of mtDNA) are recognized as markers of age-related mutations in mammals (19), and in parallel the same substitution is influenced by the metabolic levels in vertebrates, being more prevalent in species with higher body temperature (20). This sets the stage for an intriguing comparison: what would we expect when comparing, for instance, a mouse and a naked mole-rat? On one hand, the naked mole-rat might exhibit higher $T > C$ rates due to its significantly longer lifespan, but on the other hand, it might show lower $T > C$ rates due to its reduced body temperature. A quick 5-min analysis using NeMu web server to examine the mutational spectrum for CYTB reveals that $T > C$ rates are quite similar in both species (0.20 in mice and 0.21 in naked mole-rat, see the corresponding output for CYTB of the naked mole-rat https://nemu-pipeline.com/Output/?job_id=naked_mole-rat_cytb), suggesting that the two factors might balance each other out. Expanding this type of analysis to more genes and species, within the context of the relationship between life-history traits and mutational spectra, could usher in a fascinating era of research, decoding species-specific mutagens.

Limitations

Our limited testing across a random selection of genes and species indicates that approximately 10–20% of the gene-by-species samples possess the ten polymorphic sequences, enabling the NeMu pipeline to reconstruct the mutational spectrum. This proportion varies significantly across different taxa and genes. For example, we got only approximately 10 000 successful runs on 13 mtDNA protein-coding genes of Chordates species out of about 210 000 pieces available in GenBank and MIDORI2 (Supplementary Table S4). However, as the GenBank database expands, we anticipate a rise in the number of reconstructable mutational spectra, enabling the scientific community to tackle an expanding array of research questions.

Conclusion

Thus, we have developed NeMu, a flexible, convenient and interactive pipeline and web server that makes it possible to accurately disentangle neutral mutational spectra from evolutionary data. Its flexibility allows for the estimation of mutational spectra that are specific to individual species and higher

taxa. Since the neutral mutational spectrum is the focus of most evolutionary studies, the NeMu pipeline could be the software that biologists need today.

Data availability

The NeMu webserver and detailed documentation is freely available to all users at <https://nemu-pipeline.com>. The source code for a standalone version of NeMu pipeline and Singularity container, containing all dependencies, can be found at <https://github.com/mitoclub/nemu-pipeline> and <https://doi.org/10.6084/m9.figshare.25765575.v3>.

Supplementary data

Supplementary Data are available at NAR Online.

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The webserver for the NeMu pipeline is hosted on a Linux virtual server, and the backend execution is performed on the high-performance computing platform provided by Immanuel Kant Baltic Federal University.

Author contributions: The design of the study developed by K.G., the pipeline implemented by B.E. and K.G. Data mining, processing and analysis performed by B.E. and K.G. Web-interface developed by B.E. Manuscript prepared by K.G, B.E. and K.P. All authors (B.E., K.P. and K.G.) discussed in depth the manuscript and the rationale behind the project. All authors read and approved the final manuscript.

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Conflict of interest statement

None declared.

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